Batching and epochs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable batching and aggregation record.

Count schedules,repair stale gradients and weight unequal batch means correctly.

Keep one habit: W1–W3/W5 → selected new order case → project schedule ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- In Week 12 every **update** used both **training** examples. We now keep the same network h = ReLU(x), y = vh, with only v trained, and divide the data into **batch**es. In this lesson each nonempty **batch** produces exactly one **update**.

  An **epoch** visits every **training** example exactly once, without replacement. These rules make the counts unambiguous.

- Imagine five fictional records A through E in fixed order. With **batch size** 2, the groups are [A, B], [C, D] and [E]. Keep the smaller last **batch**.

  One **epoch** has three **batch**es and three **update**s; two **epoch**s have six **update**s and ten example visits.

- When the last **batch** is retained, **update**s per **epoch** equal the row count divided by **batch size** rounded up. A program that drops the final incomplete **batch** follows a different rule and omits examples in that pass.

  Record the rule instead of silently changing the denominator.

Training a small network

> **My remembered idea, example and correction**
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Gradients with a numerical check

> **My remembered idea, example and correction**

Parameters and architecture

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Training rows	Batch size	Batch lengths	Updates per epoch
5	2	2, 2, 1	3
5	5	5	1
5	1	1, 1, 1, 1, 1	5

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- One nonempty batch→one update; epoch visits every row once without replacement; retain last short batch. Five IDs, size 2→lengths 2/2/1, 3 updates; 2 epochs6 updates/10 visits. Size5 twoepochs2 updates/10 visits. Main numerical pairsA(1, 2), B(2, 4), hReLU(x), onlyvtrained, start 0, rate.1:ABgradients−4/−12.8, v.4/1.68; BAgradients−16/−.8, v1.6/1.68; fullbatchmean−10→v1.

  Main matching final values are a coincidence, not general order invariance. PictureA(1, 2), B(2, 0)instead gives AB.08, BA.4, fullbatch.2. Separate fixed squared errors[1, 1, 4, 4, 9]giveMSE19/5=3.8. Frozen sizes 3/1 means2/10 give4. Independent pairs have new targets.

- For 10 rows, **batch size** 4 and three **epoch**s, retaining the final **batch**, list **batch** lengths and count total **update**s and example visits.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- One nonempty batch→one update; epoch visits every row once without replacement; retain last short batch. Five IDs, size 2→lengths 2/2/1, 3 updates; 2 epochs6 updates/10 visits. Size5 twoepochs2 updates/10 visits. Main numerical pairsA(1, 2), B(2, 4), hReLU(x), onlyvtrained, start 0, rate.1:ABgradients−4/−12.8, v.4/1.68; BAgradients−16/−.8, v1.6/1.68; fullbatchmean−10→v1.

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- For A then B separately in the lesson, a learner computes both gradients at v = 0, then applies them to get v = 2. Explain the error and give the correct final value.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- One nonempty batch→one update; epoch visits every row once without replacement; retain last short batch. Five IDs, size 2→lengths 2/2/1, 3 updates; 2 epochs6 updates/10 visits. Size5 twoepochs2 updates/10 visits. Main numerical pairsA(1, 2), B(2, 4), hReLU(x), onlyvtrained, start 0, rate.1:ABgradients−4/−12.8, v.4/1.68; BAgradients−16/−.8, v1.6/1.68; fullbatchmean−10→v1.

  Main matching final values are a coincidence, not general order invariance. PictureA(1, 2), B(2, 0)instead gives AB.08, BA.4, fullbatch.2. Separate fixed squared errors[1, 1, 4, 4, 9]giveMSE19/5=3.8. Frozen sizes 3/1 means2/10 give4. Independent pairs have new targets.

- At one frozen model, **batch**es of sizes 3 and 1 have mean squared errors 2 and 10. Find the full MSE and explain why the average of 2 and 10 is wrong.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- One nonempty batch→one update; epoch visits every row once without replacement; retain last short batch. Five IDs, size 2→lengths 2/2/1, 3 updates; 2 epochs6 updates/10 visits. Size5 twoepochs2 updates/10 visits. Main numerical pairsA(1, 2), B(2, 4), hReLU(x), onlyvtrained, start 0, rate.1:ABgradients−4/−12.8, v.4/1.68; BAgradients−16/−.8, v1.6/1.68; fullbatchmean−10→v1.

  Main matching final values are a coincidence, not general order invariance. PictureA(1, 2), B(2, 0)instead gives AB.08, BA.4, fullbatch.2. Separate fixed squared errors[1, 1, 4, 4, 9]giveMSE19/5=3.8. Frozen sizes 3/1 means2/10 give4. Independent pairs have new targets.

- A program **process**es 11 rows using size 4 and drops the last incomplete **batch**. How many examples did it visit and how many **update**s occurred? What must its report disclose?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Test order on a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **batch**ing and **epoch**s **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish an example a **batch** an **update** and an **epoch**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Repair a stale gradient for **Batch**ing and **epoch**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Batch**ing and **epoch**s. Proposed explanation: "One **epoch** always means one **update**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use A = (1, 1), B = (2, 0), h = ReLU(x), y = vh, v = 0 and rate 0.1. Train with size 1 for one **epoch** in A then B order, then restart at 0 and reverse the order. Show gradients and parameter values. Compare the final values.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

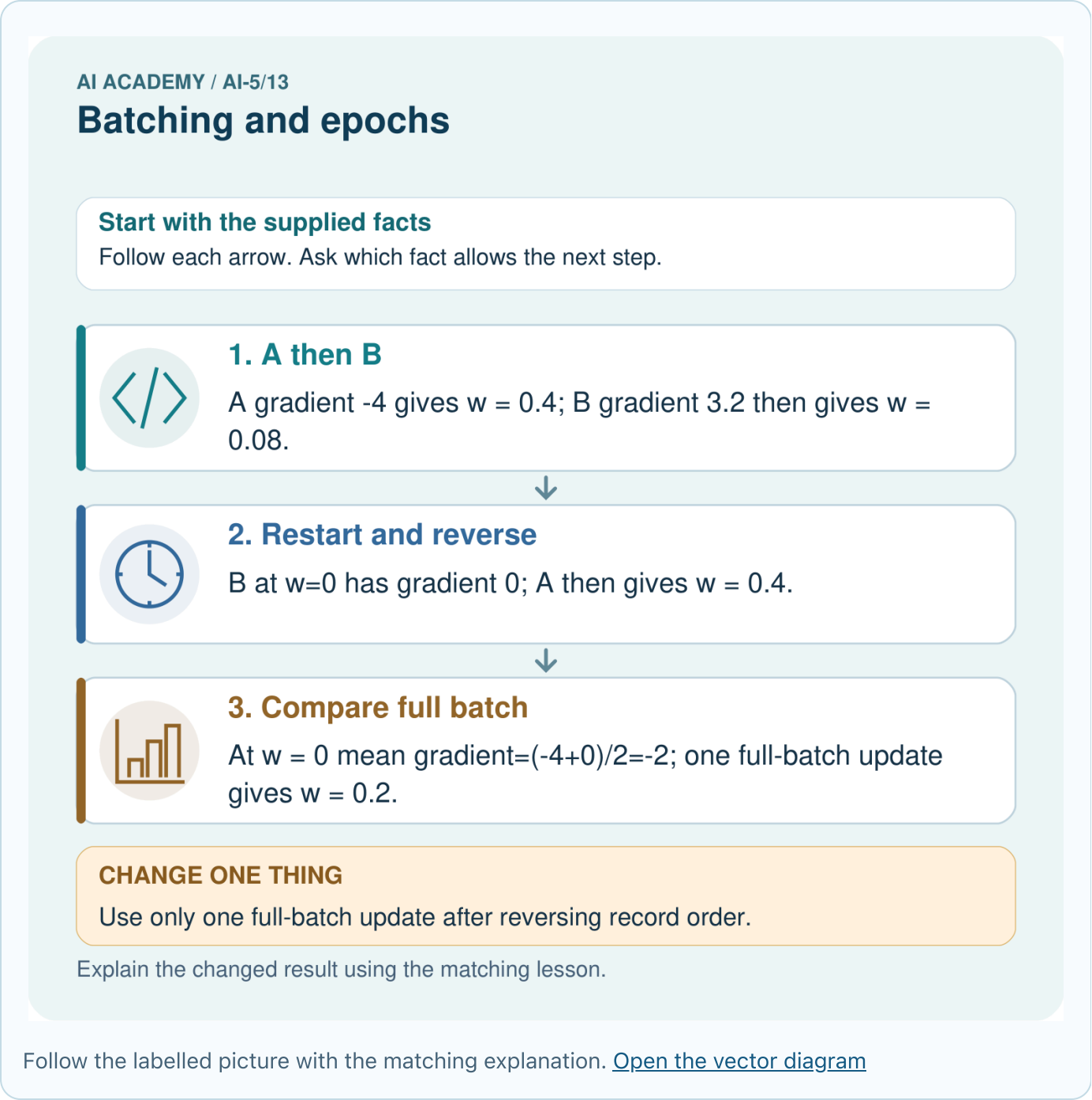
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Show why an **epoch**'s example order can matter for single-example **update**s.
- Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Show why an **epoch**'s example order can matter for single-example **update**s.
- Which labelled step matches this detail? At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.
- 1. Restart and reverse
- 2. A then B

- 3. Compare full **batch**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare full **batch** At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Changed-case question and optional hint

- Changed-case practice: Use only one full-**batch update** after reversing record order. Which conclusion is supported by the supplied facts?

- 1. The **update**d weight remains 0.2.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The **update**d weight remains 0.2. The same mean of the two starting gradients is unchanged by their order.

Repair and retain the record

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and **update** steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Batching and epochs: the case

AI ACADEMY / AI-5/13 · 01 · LOOK AT THE FACTS

Batching and epochs: the case

The situation

Show why an epoch's example order can matter for single-example updates.

Supplied evidence

Toy y=wx, rate 0.1, squared loss, start w = 0. Training A=(x = 1, target 2), B=(x = 2, target 0).
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain batching and epochs using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

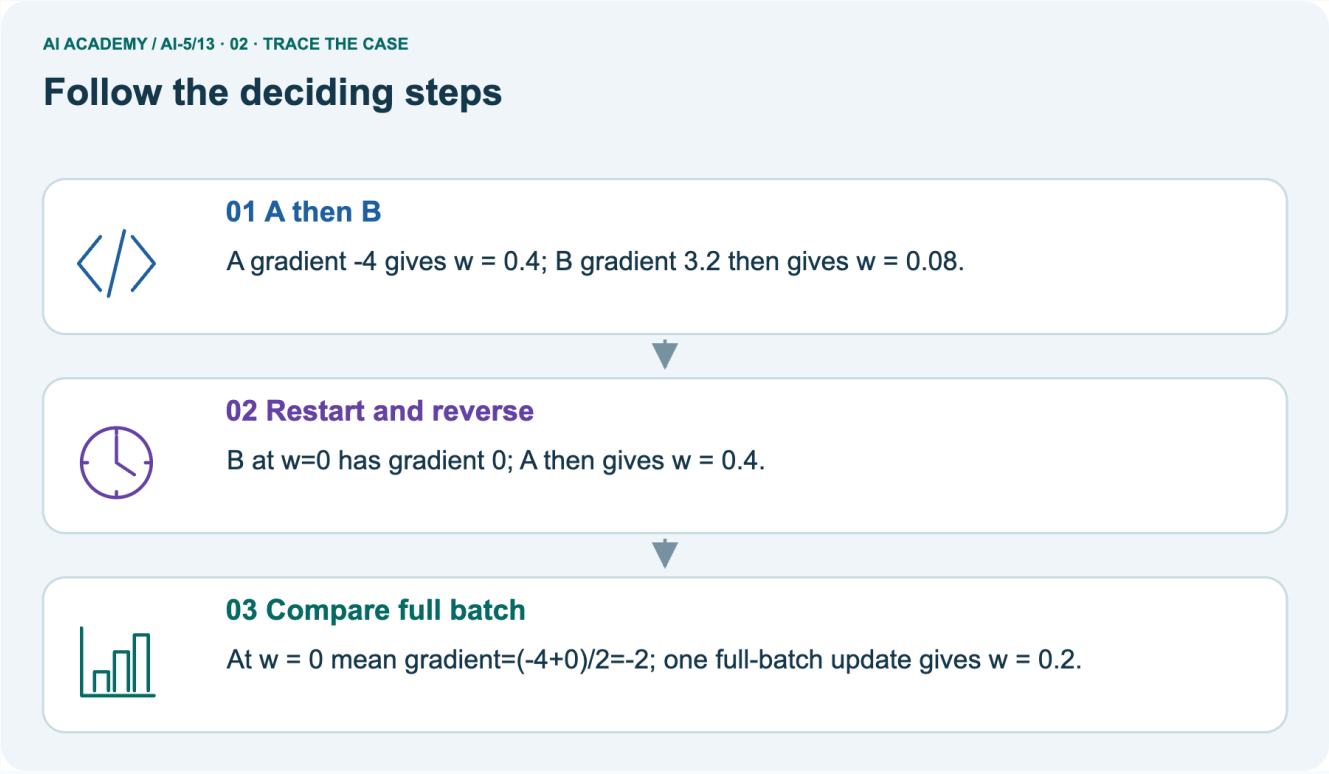
Show why an **epoch**'s example order can matter for single-example **update**s. Supplied facts: Toy y=wx, rate 0.1, squared loss, start w = 0. **Training** A=(x = 1, target 2), B=(x = 2, target 0).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

A then B: A gradient -4 gives w = 0.4; B gradient 3.2 then gives w = 0.08. → Restart and reverse: B at w=0 has gradient 0; A then gives w = 0.4. → Compare full **batch**: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One pass can contain different numbers of updates

AI ACADEMY / AI-5/13 · 03 · INSPECT THE LESSON MODEL

One pass can contain different numbers of updates

Batch rule	Parameter after each batch	Updates
A then B separately	0 to 0.4 to 1.68	2
A and B together	0 to 1	1
B then A separately	0 to 1.6 to 1.68	2

All three passes visit two examples. The reversed order happens to end at the same value for these proportional targets; that coincidence is not a general order-invariance rule.

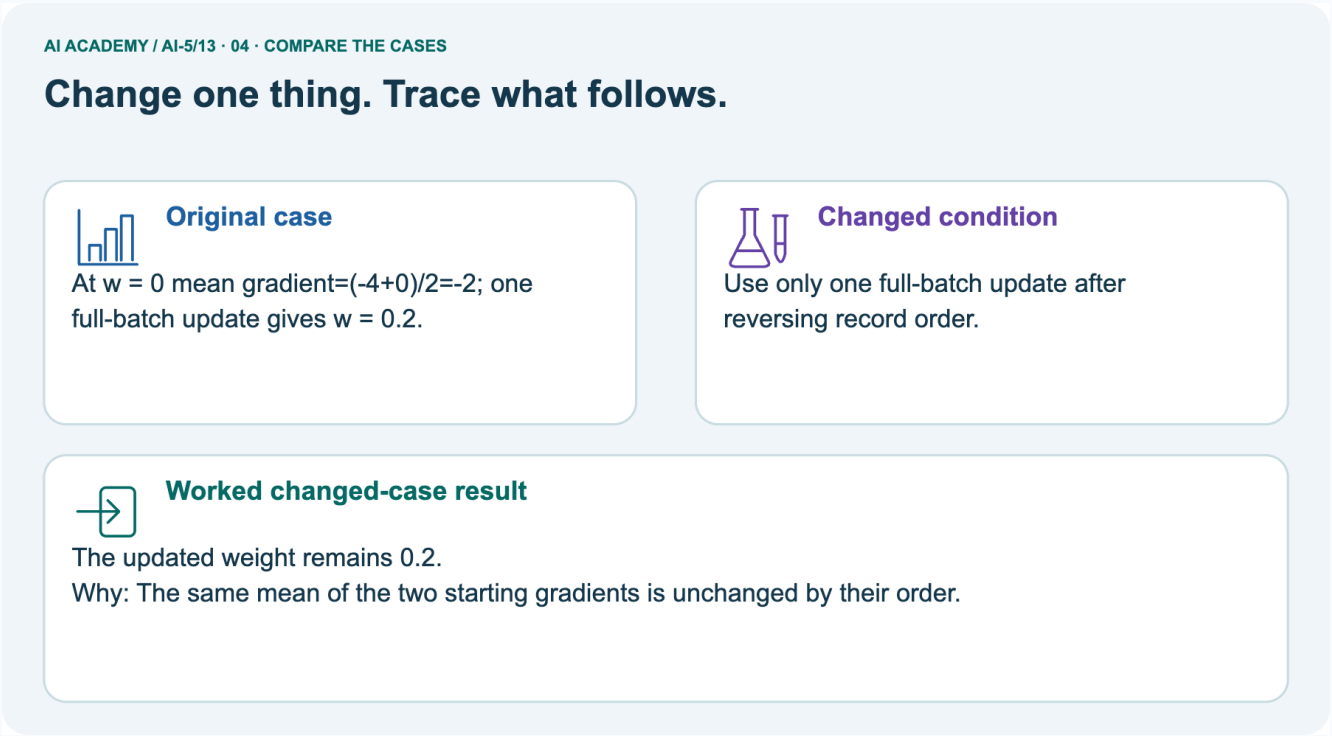
Follow the labelled picture with the matching explanation. Open the vector diagram

One pass can contain different numbers of **update**s. A then B separately | 0 to 0.4 to 1.68 | 2. A and B together | 0 to 1 | 1. B then A separately | 0 to 1.6 to 1.68 | 2

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At w = 0 mean gradient=(-4+0)/2=-2; one full-**batch update** gives w = 0.2. Changed condition: Use only one full-**batch update** after reversing record order. Expected result: The **update**d weight remains 0.2.

Reason: The same mean of the two starting gradients is unchanged by their order.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-5/13 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

A single-example update changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path. A full-batch update computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the update occurs, not whether the examples have different labels.

What it does not establish

Different order effects do not mean a universally better batching strategy; compare declared schedules and budgets fairly.

Follow the labelled picture with the matching explanation. Open the vector diagram

A single-example **update** changes the weight before the next example's gradient is calculated. Reversing the example order can therefore change the path.

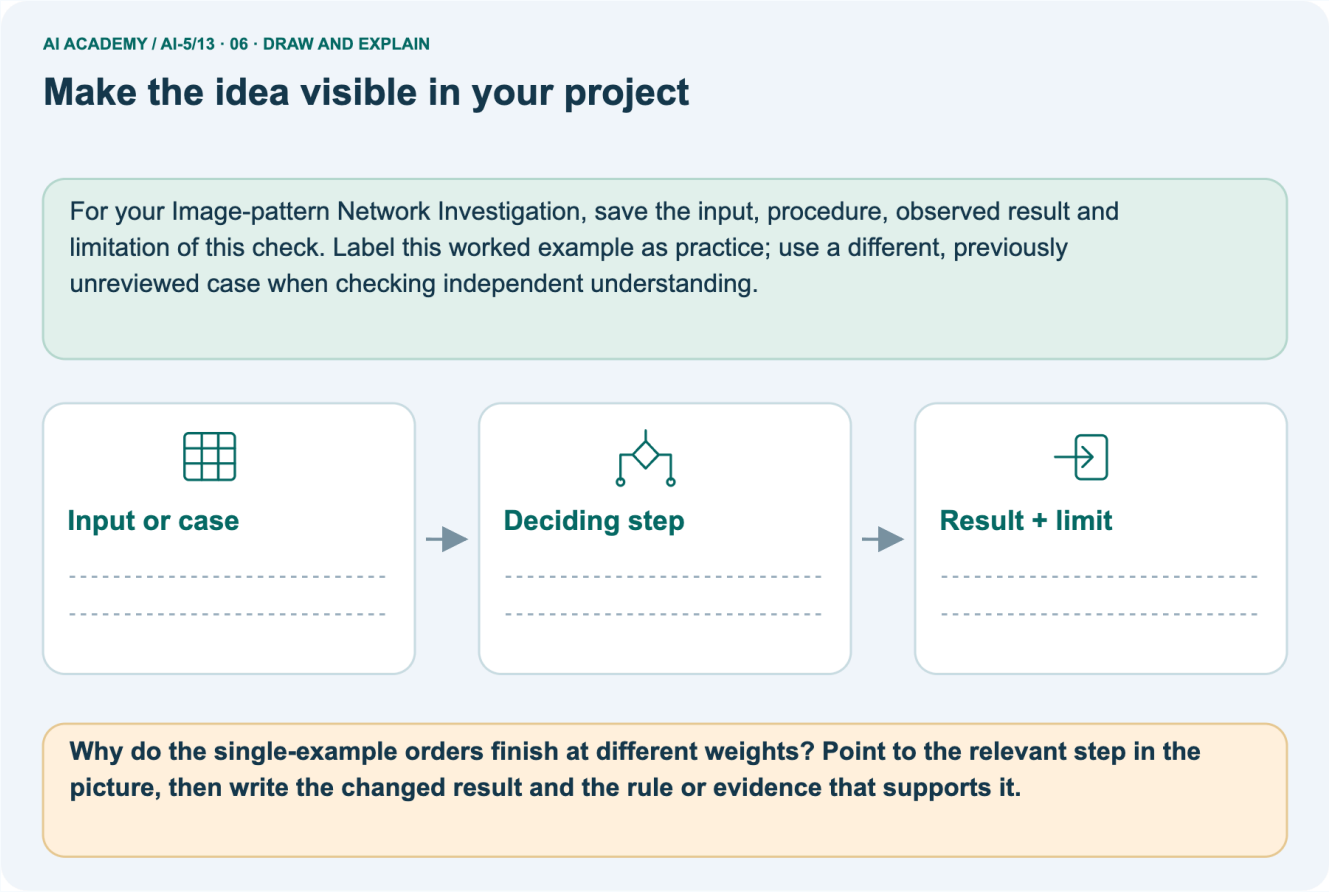
A full-**batch update** computes both gradients at the same starting weight and averages them before changing that weight. The distinction is when the **update** occurs, not whether the examples have different labels.

Limit: Different order effects do not mean a universally better **batch**ing strategy; compare declared schedules and budgets fairly.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Batching and epochs: the case

02. Follow the deciding steps

03. One pass can contain different numbers of updates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.